Practice workbook · Correlations and scatter diagrams

The correlation method and reading the direction and values of relationships between co-variables.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-RM-08

1. Question 4 · part b · 1 mark

2023 JUN · Paper 02

Worked-example retry · Attempt part b only.

4 Research in psychology can use a correlation research method.

(a) Define what is meant by a 'correlation research method'.

(1)

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A scatter diagram to show the correlation between maths end of topic test scores and the number of homework tasks missed

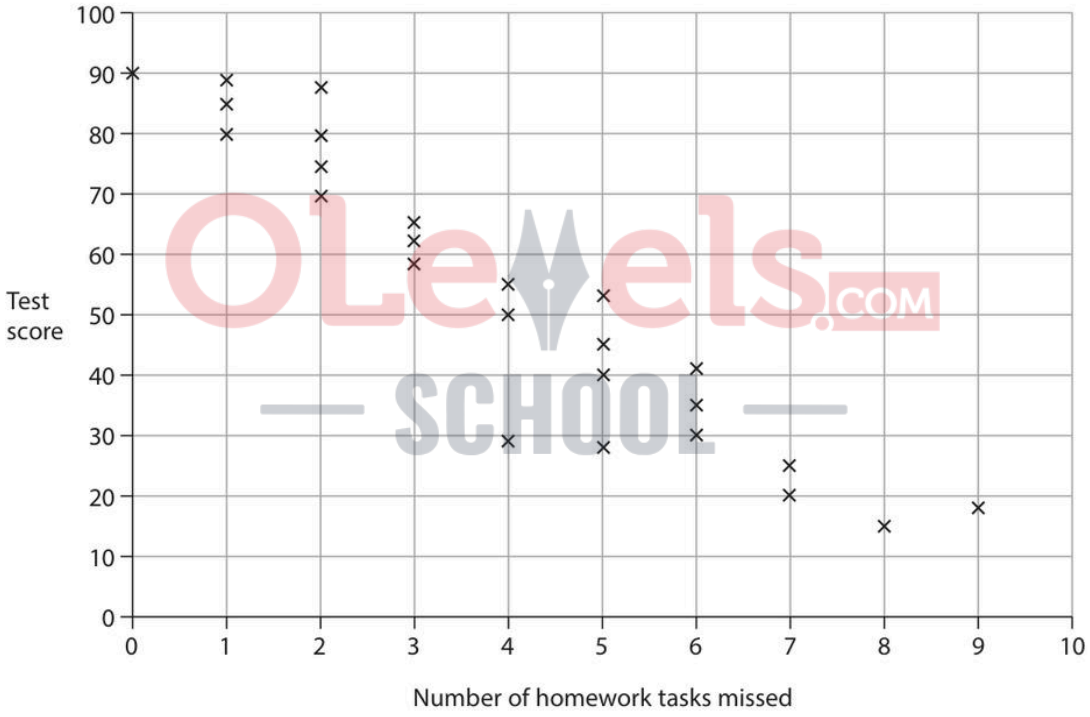


Figure 2

(b) Identify the type of correlation shown in **Figure 2**.

(1)

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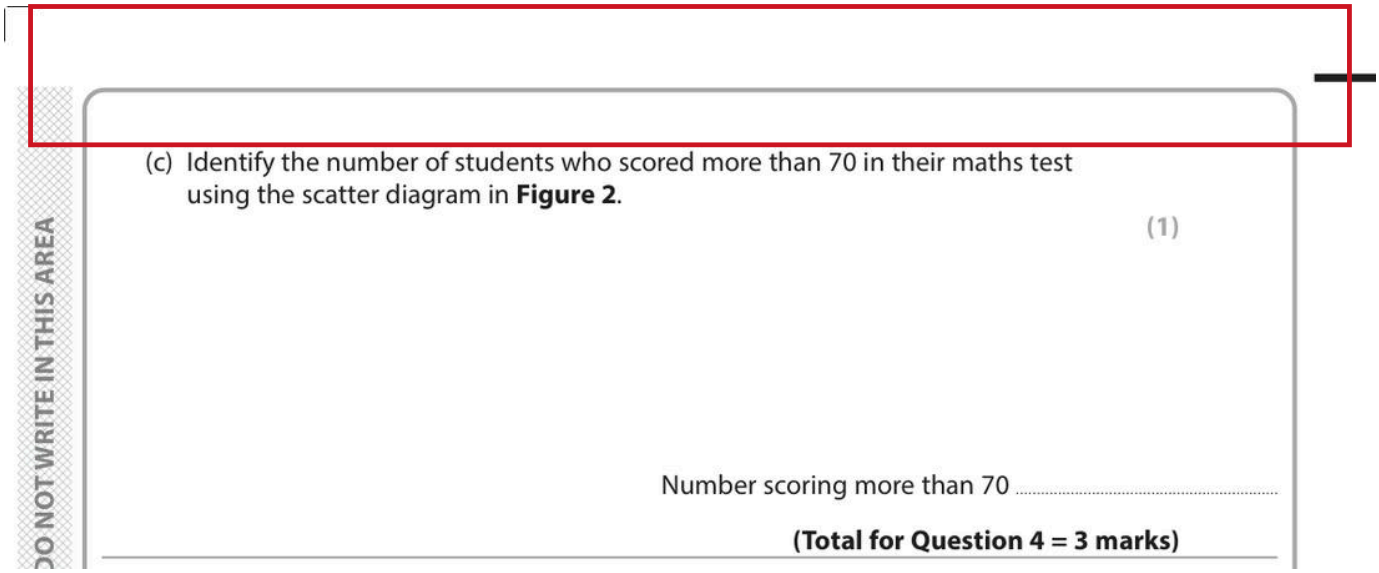
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P 7 2 3 8 3 A 0 1 0 4 8

1. Question 4 · part b · 1 mark

2023 JUN · Paper 02

Worked-example retry · Attempt part b only.



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(c) Identify the number of students who scored more than 70 in their maths test using the scatter diagram in **Figure 2**.

(1)

Number scoring more than 70 .....

(Total for Question 4 = 3 marks)

Working and answer

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2. Question 3 · part e · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part e only.

3 Juliet investigated whether there was a relationship between the number of hours spent using social media and self-rated body image scores.

Firstly, Juliet interviewed a sample of 17-year-old students using a structured interview method, where she asked them to state the length of time they spent on social media per day.

Juliet then asked the same participants to complete a questionnaire to rate how happy they were with their appearance, with a score of 0 being 'very unhappy' and 10 being 'very happy'.

(a) Define what is meant by a 'structured interview'.

(1)

(b) Describe how Juliet could have used a random sampling technique for her investigation.

(2)

(c) Identify the type of data Juliet gathered in her investigation.

(1)

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2. Question 3 · part e · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part e only.

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The results of Juliet's investigation are shown in **Table 3**.

Participant	Number of hours per day using social media	Body image score (out of 10)
A	2	6
B	1	7
C	4	3
D	6	2
E	2	7
F	5	3
G	1	9
H	2	7
I	7	1

Table 3

(d) Calculate the mode for body image scores.

(1)

Mode .....

(e) Juliet concludes that there is a negative relationship between social media use and body image.

Give **one** way Juliet's findings could support this conclusion.

(1)

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(Total for Question 3 = 6 marks)

2. Question 3 · part e · 1 mark · Working and answer

Working and answer

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3. Question 4 · part a · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part a only.

4 Research in psychology can use a correlation research method.

(a) Define what is meant by a 'correlation research method'.

(1)

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A scatter diagram to show the correlation between maths end of topic test scores and the number of homework tasks missed

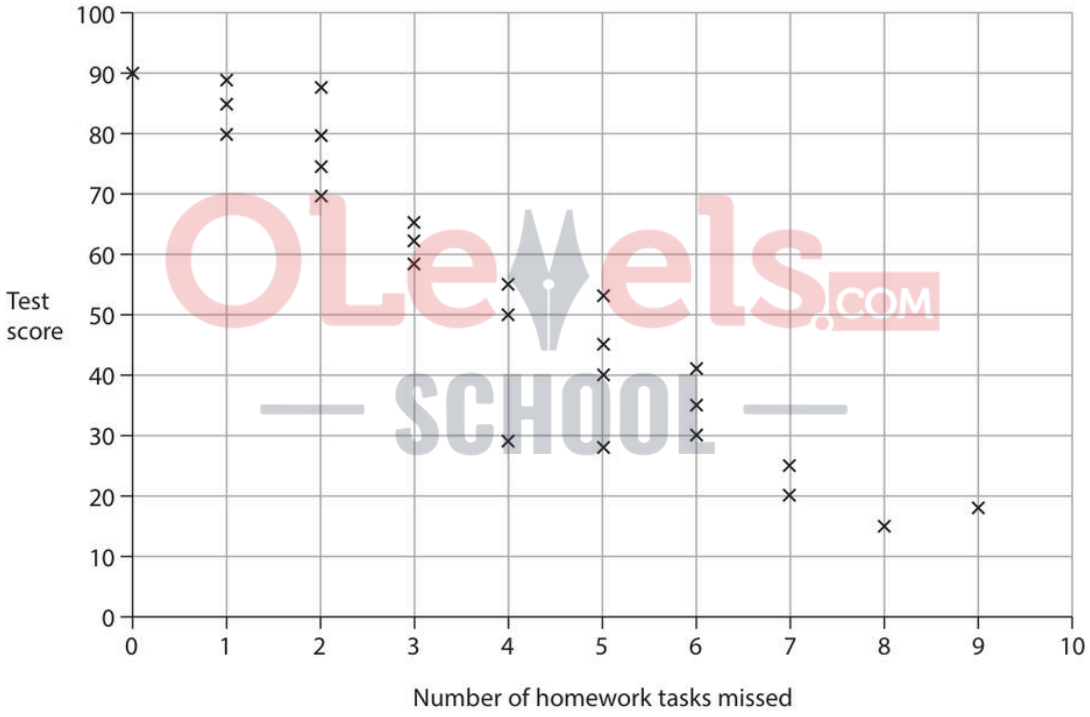


Figure 2

(b) Identify the type of correlation shown in **Figure 2**.

(1)

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3. Question 4 · part a · 1 mark · Working and answer

Working and answer

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4. Question 4 · part c · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c only.

4 Research in psychology can use a correlation research method.

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A scatter diagram to show the correlation between maths end of topic test scores and the number of homework tasks missed

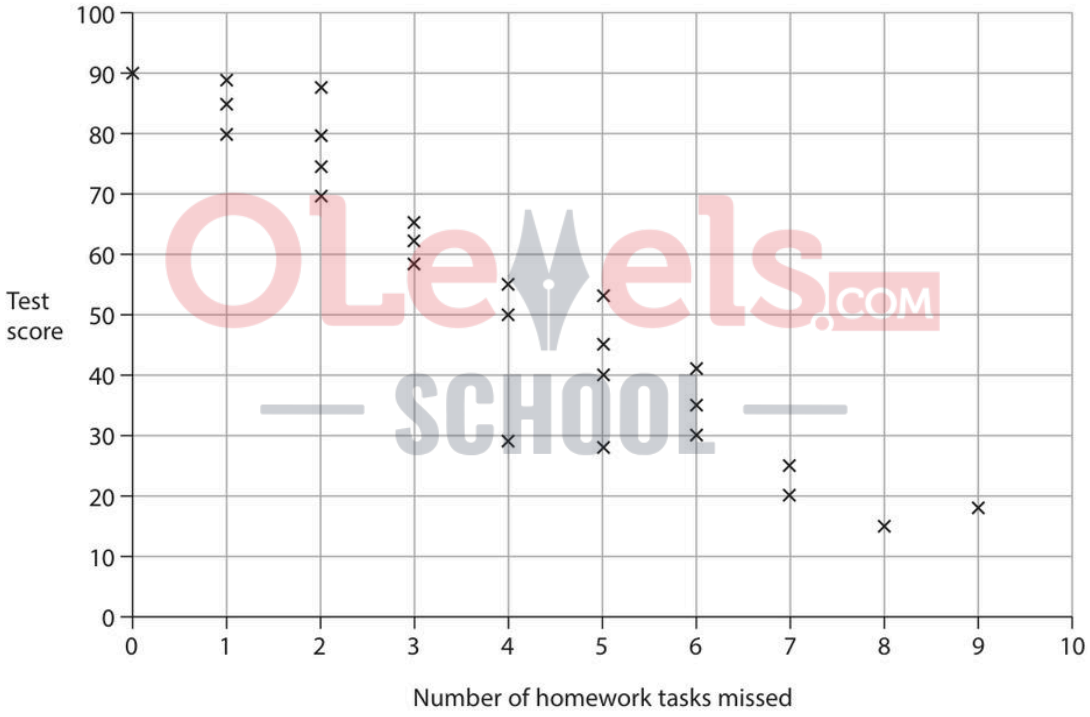


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4. Question 4 · part c · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c only.

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(c) Identify the number of students who scored more than 70 in their maths test using the scatter diagram in **Figure 2**.

(1)

Number scoring more than 70 .....

(Total for Question 4 = 3 marks)

Working and answer

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5. Question 3 · part c · 2 marks

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Further practice · Attempt part c only.

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3 Savannah is investigating whether there is a relationship between listening to music and happiness. She asked participants to record the number of hours they spent listening to music in a week. They also scored their level of happiness on a scale of 1 (being very unhappy) to 10 (being very happy).

Savannah sampled a total of 180 participants aged between 18 years old and 65 years old. A third of her sample were aged between 18 years old and 25 years old.

(a) Calculate the range for the age of all participants.

(1)

Range .....

(b) Calculate the ratio of participants aged between 18 years old and 25 years old to the remaining participants in the sample used by Savannah.

(2)

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Ratio .....

(c) Savannah found that those listening to less than one hour of music in a week scored themselves at 2 or below on the happiness scale. Happiness scores increased for those who listened to more music, with those who listened to over 12 hours of music in a week usually scoring 9 or above.

Explain **one** conclusion that Savannah could make from her findings about the relationship between music and happiness.

(2)

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(Total for Question 3 = 5 marks)

5. Question 3 · part c · 2 marks · Working and answer

Working and answer

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